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/ [Test 3 \(topics 7-9: Recurrence relations, Introduction to group theory, The multiplicative group of integers mod\(n\) and Lagrange's theorem\)](#)

Started on Wednesday, 6 November 2024, 1:45 PM

State Finished

Completed on Wednesday, 6 November 2024, 1:57 PM

Time taken 11 mins 56 secs

Marks 3.00/3.00

Grade 3.00 out of 3.00 (100%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade**.

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question **1**

Correct

Mark 1.00 out of 1.00

Find the solution to the initial value problem

$$\begin{cases} a_n - 4 \cdot a_{n-1} + 4 \cdot a_{n-2} = 0 \\ a_0 = 1 \\ a_1 = 10 \end{cases}$$

$a_n =$

Input your answer as a function of n . Any other variable is treated as a constant. E.g. if you think the answer is $2^n + 3$, you should enter **$2^n + 3$** .

Your last answer was interpreted as follows:

$$(1 + 4 \cdot n) \cdot 2^n$$

The variables found in your answer were: $[n]$

Question **2**

Correct

Mark 1.00 out of 1.00

In this problem you are tasked with computing subgroups of $(\mathbb{Z}/n, +_n)$ and direct products of such groups.

a) Find the subgroup generated by the element $0 \in \mathbb{Z}/3$.

$\langle 0 \rangle =$

Your last answer was interpreted as follows:

$$\{0\}$$

Syntax guide: Input your answer as a set. E.g. if you think the answer is $\{1, 2, 3\}$, then you should type **{1,2,3}**.

b) Find the subgroup generated by the element $2 \in \mathbb{Z}/5$.

$\langle 2 \rangle =$

Your last answer was interpreted as follows:

$$\{0, 1, 2, 3, 4\}$$

Syntax guide: Input your answer as a set. E.g. if you think the answer is $\{1, 2, 3\}$, then you should type **{1,2,3}**.

c) Find the subgroup generated by the element $[0, 2] \in \mathbb{Z}/3 \times \mathbb{Z}/5$.

$\langle [0, 2] \rangle =$

Your last answer was interpreted as follows:

$$\{[0, 0], [0, 2], [0, 4], [0, 1], [0, 3]\}$$

Syntax guide: Elements of this group are on the form $[a, b]$. Your answer should be a set of such elements. E.g. if you think the answer is $\{[1, 2], [0, 2], [1, 3]\}$, then you should type **{[1,2],[0,2],[1,3]}**.

Question 3

Correct

Mark 1.00 out of 1.00

Let $(P_9, \bullet_9)$ be the group of natural numbers coprime to 9 with group operation given by multiplication modulo 9. The multiplication table for this group looks like

1	2	4	5	7	8
2	4	8	.	.	.
4	8	7	2	1	5
5	1	2	.	.	.
7	5	1	8	4	2
8	7	5	.	.	.

However, we have left out 9 entries in the places marked by a dot "."

Compute the missing entries in the group multiplication table:

$5 \bullet_9 7 =$	8
$8 \bullet_9 7 =$	2
$2 \bullet_9 7 =$	5
$5 \bullet_9 5 =$	7
$8 \bullet_9 5 =$	4
$2 \bullet_9 5 =$	1
$5 \bullet_9 8 =$	4
$8 \bullet_9 8 =$	1
$2 \bullet_9 8 =$	7

Your last answer was interpreted as follows:

8

Your last answer was interpreted as follows:

2

Your last answer was interpreted as follows:

5

Your last answer was interpreted as follows:

7

Your last answer was interpreted as follows:

4

Your last answer was interpreted as follows:

1

Your last answer was interpreted as follows:

4

Your last answer was interpreted as follows:

1

Your last answer was interpreted as follows:

7

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 541

Signing code:

Your last answer was interpreted as follows:

17885

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

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